

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date:	29/09/26	Team ID:	SWTID-2026-8956
Project Name:	WeatherWise AI	Maximum Marks:	4 Marks
Team Size:	4 Members		
Team Members:	SANMATHI KUMAR (Leader), ARIFA ZOHRA, AINUL LIYANA P M, VAISHALI S		

1. FUNCTIONAL REQUIREMENTS (FR)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Registration	- Registration via Email & Password (<code>/api/auth/register</code>) - Login authentication via password verification (<code>/api/auth/login</code>) - Password security using <code>bcryptjs</code> hashing - JWT Middleware for protected route authorization
FR-2	Saved Locations Management	- Add favorite city/country mapped to User ObjectId - Fetch list of user's saved locations (<code>GET /api/locations</code>) - Delete location (<code>DELETE /api/locations/:id</code>) - Enforce compound index <code>{ user: 1, city: 1, country: 1 }</code>
FR-3	Weather Data Service	- Fetch live weather metrics via OpenWeatherMap API - Parse temperature, humidity, wind speed, and conditions - Expose guest endpoints for non-authenticated lookup
FR-4	Gemini AI Recommendation Engine	- Integrate <code>@google/genai</code> generative AI SDK - Generate activity/attire suggestions (<code>POST /api/ai/weather-recommendation</code>) - Automated Fallback Mode if API keys are absent

2. NON-FUNCTIONAL REQUIREMENTS (NFR)

NFR No.	Requirement	Description
NFR-1	Usability	RESTful interface producing standardized JSON payloads for clients.
NFR-2	Security	JWT session protection, bcryptjs password hashing, and input sanitization to block XSS and NoSQL injections.
NFR-3	Reliability	Built-in Fallback Mode ensuring operational integrity during third-party API downtime.
NFR-4	Performance	Asynchronous I/O ensuring <500ms core route latency and sub-2s AI output generation.
NFR-5	Availability	Designed for 99.9% availability via cloud service deployment.
NFR-6	Scalability	Decoupled MVC architecture supporting horizontal scaling.